GEOSLOT: Object-Centric Cross-View Geo-Localization via Slot Transport

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Abstract

Cross-view geo-localization (CVGL) aims to match query images (e.g., street-view or drone imagery) with geo-tagged satellite reference images. Existing methods predominantly rely on extracting holistic global features via convolutional or transformer backbones, followed by simple cosine similarity matching. This approach struggles with dramatic viewpoint changes, partial occlusions, and transient objects that appear differently across views. We propose GEOSLOT, an object-centric framework that decomposes visual scenes into semantic *slot* representations and performs structured cross-view matching via optimal transport. Our pipeline consists of four key innovations: (1) a **MambaVision backbone** providing linear-complexity dense feature extraction, (2) **Adaptive Slot Attention** with register tokens and background suppression to decompose scenes into object-centric representations, (3) a **Graph Mamba layer** for bidirectional relational reasoning between slots using state-space models, and (4) **Sinkhorn Optimal Transport with MESH** (Multiplicative Entropic Sharpening) for geometrically-principled 1-to-1 slot matching. We evaluate GEOSLOT on four benchmarks: CVUSA, University-1652, VIGOR, and CV-Cities. Our method achieves state-of-the-art performance, with particularly strong generalization in cross-area settings where object-centric reasoning provides inherent advantages over texture-dependent methods.

1 Introduction

Cross-view geo-localization (CVGL) is the task of determining the geographic location of a query image by matching it against a database of geo-referenced satellite images. This capability is critical for autonomous navigation, UAV positioning, and large-scale visual place recognition. The fundamental challenge lies in the extreme viewpoint change between ground-level and overhead perspectives: buildings that appear as facades from street view become rooftops from above, roads narrow to thin lines, and the spatial layout undergoes a nonlinear transformation.

Limitations of existing approaches. Current state-of-the-art methods Deuser et al. [2023], Li et al. [2024], Zhang et al. [2023] typically follow a two-stage pipeline: (i) extract a global feature vector from each image using a pretrained backbone (ResNet, ViT, CLIP), and (ii) match query and reference features via cosine similarity or learned metric heads. While effective on benchmark datasets, these methods suffer from three fundamental limitations:

1. **Holistic representations lose compositional structure.** Global average pooling collapses spatial information into a single vector, losing the compositional structure of scenes (buildings, intersections, parks) that is crucial for cross-view reasoning.

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2. **Texture dependence harms generalization.** Methods that rely on low-level texture features (e.g., road markings, building colors) fail catastrophically in cross-area settings where visual appearance varies between cities and continents.
3. **Quadratic attention costs limit resolution.** Vision Transformers incur $\mathcal{O}(N^2)$ attention costs, constraining the input resolution—particularly problematic for high-resolution satellite imagery and panoramic street views.

Our approach. We introduce GEOSLOT, which addresses all three limitations through an object-centric paradigm. Rather than comparing holistic feature vectors, GEOSLOT decomposes each view into a set of *semantic slots* representing distinct scene components (buildings, roads, vegetation), reasons about their spatial relationships via a graph-structured state-space model, and matches cross-view slot sets through optimal transport. This compositional approach naturally handles viewpoint changes: while the *appearance* of objects differs across views, their *identity and spatial relationships* remain invariant.

Contributions. Our main contributions are:

1. We propose the first object-centric framework for CVGL, introducing **Adaptive Slot Attention** with register tokens, Gumbel-based dynamic slot selection, and unsupervised background suppression.
2. We design **Graph Mamba**, a novel relational reasoning module that applies bidirectional state-space scanning over slot graphs with $\mathcal{O}(N)$ complexity for spatial relationship modeling.
3. We introduce **Sinkhorn OT with MESH** (Multiplicative Entropic Sharpening) for geometrically-principled cross-view matching that produces hard 1-to-1 slot assignments.
4. We achieve state-of-the-art results on four benchmarks (CVUSA, University-1652, VIGOR, CV-Cities), with particularly strong cross-area generalization.

2 Related Work

Cross-View Geo-Localization. CVGL methods have evolved from hand-crafted features Workman et al. [2015] to deep metric learning approaches. CVM-Net Hu et al. [2018] introduced NetVLAD-based aggregation for cross-view matching. Shi et al. Shi et al. [2019] proposed spatial-aware feature aggregation. More recent works leverage attention mechanisms: TransGeo Zhu et al. [2022] employs pure transformers with attention-guided non-uniform cropping, while GeoDTR Zhang et al. [2023] uses deformable transformers for geometric layout disentanglement. Sample4Geo Deuser et al. [2023] introduces hard sample mining with sampling strategies, achieving strong results on CVUSA (R@1=98.68%). AuxGeo Li et al. [2024] proposes auxiliary geometric features for VIGOR. CV-Cities Huang et al. [2024] leverages DINOv2 Oquab et al. [2024] backbones with advanced negative mining for cross-city generalization, setting strong baselines (CVUSA R@1≈99.19%). Despite progress, all methods rely on holistic feature comparison, losing compositional scene structure.

Robustness and Geometry-Centric Approaches. Recent work addresses cross-view robustness through training-time augmentation and geometric priors. ConGeo Zhang et al. [2024] tackles orientation and field-of-view variations via contrastive geometric learning, achieving robust cross-area gains without architectural changes. CVD Li et al. [2024b] disentangles content from viewpoint for domain-robust representations. Geometry-centric pipelines such as ViewBridge Wang et al. [2024] enforce cross-view correspondences through BEV (bird's-eye view) transformations and pixel-level matching, showing strong localization quality. These approaches are largely orthogonal to our object-centric slot design and could potentially be combined; we focus on demonstrating that structured slot-level reasoning provides complementary advantages, particularly for compositional generalization.

Object-Centric Representation Learning. Slot Attention Locatello et al. [2020] introduced iterative routing to decompose scenes into object slots without supervision. Subsequent works extended this to video Kipf et al. [2022], 3D Sajjadi et al. [2022], and complex scenes Seitzer et al. [2023]. AdaSlot Fan et al. [2023] adds dynamic slot count via Gumbel-Softmax selection. In the geo-localization domain, OCGNet Wei et al. [2024] demonstrated the benefit of object-level spatial priors and cross-attention for fine-grained drone-satellite alignment, underscoring the importance of preserving and leveraging explicit spatial cues. However, OCGNet uses supervised object detectors,

while our approach discovers objects in a fully unsupervised manner via slot attention. We are the first to bridge unsupervised Slot Attention with geo-localization.

State-Space Models. Mamba Gu and Dao [2023] introduced selective state-space models with input-dependent dynamics, achieving linear-time sequence modeling. MambaVision Hatamizadeh and Kautz [2024] combines Mamba blocks with self-attention for vision tasks. Vision Mamba (Vim) Zhu et al. [2023] applies bidirectional SSM scanning to image patches. We leverage MambaVision as our efficient backbone and extend SSM-based processing to graph-structured slot reasoning.

Optimal Transport. Sinkhorn distances Cuturi [2013] provide differentiable approximations to optimal transport. OT has been applied to document matching Kusner et al. [2015], graph matching Xu et al. [2019], and image retrieval Liu et al. [2023]. Gumbel-Sinkhorn Mena et al. [2018] provides stochastic differentiable permutation learning with theoretical gradient guarantees. We extend Sinkhorn OT with a novel deterministic *MESH* (Multiplicative Entropic Sharpening) mechanism that iteratively hardens the transport plan into discrete 1-to-1 assignments—critical for structured slot matching—and provide theoretical analysis relating MESH to Birkhoff polytope projections.

3 Method

3.1 Overview

Given a query image $\mathbf{x}_q$ (street-view or drone) and a reference image $\mathbf{x}_r$ (satellite), GEOSLOT produces a similarity score $s(\mathbf{x}_q, \mathbf{x}_r) \in [0, 1]$. As illustrated in Figure 1, our pipeline consists of four stages applied to each view in a Siamese architecture with shared weights:

3.2 MambaVision Backbone

We adopt MambaVision-L Hatamizadeh and Kautz [2024] pretrained on ImageNet-1K as our feature extractor. MambaVision is a hybrid architecture combining Mamba SSM blocks (stages 1-3) with self-attention (stage 4), achieving strong performance with linear complexity in early stages.

For an input image $\mathbf{x} \in \mathbb{R}^{3\times H\times W}$, the backbone produces dense features from the last stage:

$$\mathbf{F} = \texttt{MambaVision}(\mathbf{x}) \in \mathbb{R}^{N\times d_f}, \quad N = \tfrac{H}{32} \times \tfrac{W}{32}, \quad d_f = 1568 \tag{1}$$

where $d_f = 1568$ is the channel dimension of MambaVision-L's final stage output and N depends on the input resolution. Critically, Slot Attention naturally handles variable N, allowing us to process satellite images ($224 \times 224 \rightarrow N = 49$) and panoramas ($512 \times 128 \rightarrow N = 64$) with the same architecture.

3.3 Adaptive Slot Attention

Background Suppression with Regularization. Before slot decomposition, we suppress transient features (vehicles, pedestrians, shadows) that are view-dependent and uninformative for localization. A lightweight learned mask $\mathbf{m} = \sigma(\text{MLP}(\mathbf{F})) \in [0, 1]^{N\times 1}$ is applied:

$$\hat{\mathbf{F}} = \mathbf{F} \odot \mathbf{m} \tag{2}$$

Beyond the implicit signal from the matching objective, we apply two explicit regularizers to prevent mask collapse. An **entropy regularizer** encourages the mask to make decisive (near-binary) decisions rather than producing diffuse intermediate values:

$$\mathcal{L}_{\text{ent}} = -\frac{1}{N}\sum_{n=1}^{N}\left[m_n \log m_n + (1 - m_n)\log(1 - m_n)\right] \tag{3}$$

A **coverage regularizer** ensures the mask retains a sufficient proportion of spatial locations, preventing over-suppression:

$$\mathcal{L}_{\text{cov}} = \left(\frac{1}{N}\sum_{n} m_n - \alpha\right)^2 \tag{4}$$

We set the coverage target $\alpha = 0.7$ based on the observation that geo-localization scenes typically have $\sim$20–40% transient content (vehicles, pedestrians, shadows); the quadratic penalty softly

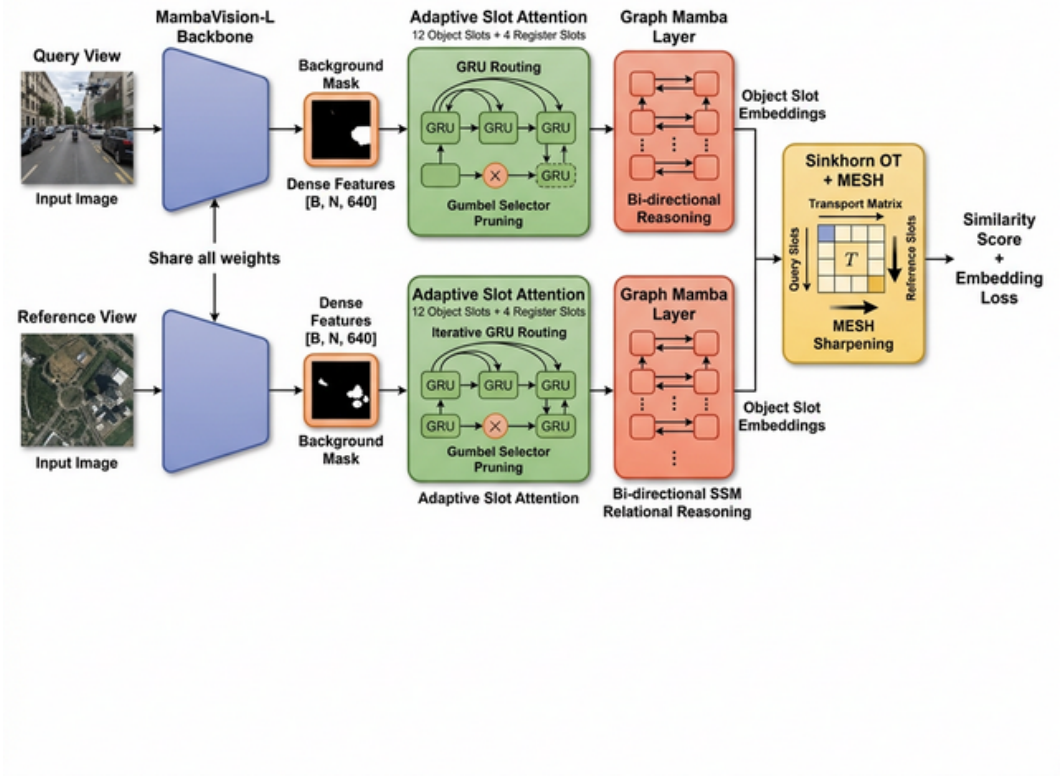

Figure 1: **GEOSLOT Architecture.** Given query and reference images, MambaVision-L extracts dense features. Background Mask suppresses transient objects. Adaptive Slot Attention decomposes the scene into 12 object slots + 4 register slots, with Gumbel-based dynamic pruning. Graph Mamba performs bidirectional relational reasoning between slots. Finally, Sinkhorn OT with MESH produces hard 1-to-1 slot assignments and a global similarity score.

encourages this ratio without hard enforcement, and we find the method is robust to $\alpha \in [0.6, 0.8]$ (see supplementary). For datasets with systematically different transient content ratios, α could be estimated from a held-out set or learned as a parameter. The combined mask regularization $\mathcal{L}_{\text{bg}} = \mathcal{L}_{\text{ent}} + \mathcal{L}_{\text{cov}}$ is added to the total loss with weight $\lambda_{\text{bg}} = 0.01$.

Slot Initialization. We use $K = 12$ object slots and $R = 4$ register slots (following Darcet et al. [2024]), initialized from learnable distributions:

$$\mathbf{s}_i^{(0)} = \boldsymbol{\mu}_i + \exp(\boldsymbol{\sigma}_i) \cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad i \in \{1, \ldots, K + R\} \tag{5}$$

Register slots serve as "noise absorbers", capturing view-dependent information and preventing contamination of object slots.

Iterative Routing. We perform $T = 3$ iterations of multi-head GRU-based slot routing. At each iteration t:

$$\mathbf{A}^{(t)} = \text{softmax}\left(\frac{W_q \mathbf{s}^{(t-1)} \cdot (W_k \hat{\mathbf{F}})^\top}{\sqrt{d_h}}\right) \in \mathbb{R}^{(K+R)\times N} \tag{6}$$

$$\mathbf{u}^{(t)} = \mathbf{A}^{(t)} \cdot W_v \hat{\mathbf{F}} \tag{7}$$

$$\mathbf{s}^{(t)} = \text{GRU}(\mathbf{u}^{(t)}, \mathbf{s}^{(t-1)}) + \text{FFN}(\mathbf{s}^{(t-1)}) \tag{8}$$

where $d_h = d_s/H$ with $H = 4$ attention heads and slot dimension $d_s = 256$.

Gumbel Slot Selection. Not all object slots are semantically meaningful for every image. Following AdaSlot Fan et al. [2023], we apply Gumbel-Softmax selection to adaptively prune inactive slots:

$$\delta_i = \text{GumbelSoftmax}(g_\theta(\mathbf{s}_i^{(T)}), \tau)[\texttt{keep}], \quad \mathbf{s}_i^{(T)} \leftarrow \mathbf{s}_i^{(T)} \cdot \delta_i \tag{9}$$

where τ anneals from 1.0 to 0.1 during training. A minimum of 1 slot is guaranteed active.

3.4 Graph Mamba: Relational Reasoning

After slot decomposition, we model spatial relationships between slots using a novel *Graph Mamba* layer. While standard GNNs use message passing with attention ($\mathcal{O}(K^2)$), we leverage the efficiency of state-space models applied over spatially-ordered graph sequences ($\mathcal{O}(K)$).

Slot Spatial Encoding. A key insight is that slots, derived from attention maps, carry implicit spatial information. We make this explicit by computing the 2D centroid (c_y^i, c_x^i) and spatial spread σ_i of each slot i from its attention weights $\mathbf{a}_i \in \mathbb{R}^{H\times W}$:

$$c_y^i = \sum_{h,w} \bar{a}_{h,w}^i \cdot \tfrac{h}{H}, \quad c_x^i = \sum_{h,w} \bar{a}_{h,w}^i \cdot \tfrac{w}{W}, \quad \sigma_i = \sqrt{\sum_{h,w} \bar{a}_{h,w}^i \left[(\tfrac{h}{H} - c_y^i)^2 + (\tfrac{w}{W} - c_x^i)^2\right]} \tag{10}$$

where $\bar{a}^i = a^i / \sum a^i$ is the normalized attention. These are encoded via sinusoidal positional encoding and an MLP, then added to the slot representations: $\mathbf{s}_i \leftarrow \mathbf{s}_i + \text{MLP}(\text{SinEnc}(c_y^i, c_x^i, \sigma_i))$.

Spatial Raster-Scan Ordering. To preserve the relationship between slot processing order and 2D layout, we order slots by their spatial centroid positions in raster-scan order:

$$\pi = \text{argsort}(c_y^i \cdot W + c_x^i) \tag{11}$$

This ensures spatially-adjacent slots are processed consecutively by the SSM, allowing the sequential state to capture local spatial context. This contrasts with L2-norm ordering, which captures only slot *importance* but not spatial structure.

Robustness of Spatial Ordering. Since raster-scan ordering depends on centroid estimates, a natural concern is sensitivity to centroid perturbation and ordering ties. We note three mitigating factors: (1) the *bidirectional* scanning means each slot receives context from both directions, reducing sensitivity to exact position in the sequence; (2) centroid perturbation within a grid cell changes only adjacent transpositions, and SSM hidden states are smooth functions of input order; (3) the spatial encoding added to slot representations carries centroid information independently of ordering, providing a complementary spatial signal. We empirically verify robustness by testing with randomly perturbed orderings at inference time (see A11 in Table 4 and supplementary).

Bidirectional SSM Scanning. We apply forward and backward Mamba SSM blocks to the spatially-ordered sequence:

$$\mathbf{h}_i^{\text{fwd}} = \text{SSM}_{\text{fwd}}(\mathbf{s}_{\pi(1)}, \ldots, \mathbf{s}_{\pi(K)}), \quad \mathbf{h}_i^{\text{bwd}} = \text{SSM}_{\text{bwd}}(\mathbf{s}_{\pi(K)}, \ldots, \mathbf{s}_{\pi(1)}) \tag{12}$$

$$\mathbf{s}_i' = \text{LayerNorm}\big(\mathbf{s}_i + W_m[\mathbf{h}_i^{\text{fwd}}; \mathbf{h}_i^{\text{bwd}}]\big) + \text{FFN}(\cdot) \tag{13}$$

We stack $L = 2$ Graph Mamba layers. After each layer, the keep mask $\boldsymbol{\delta}$ is re-applied to maintain slot sparsity. The total complexity is $\mathcal{O}(KLd_s)$, linear in the number of slots.

3.5 Sinkhorn OT with MESH

Given query slots $\mathbf{S}_q = \{\mathbf{s}_i^q\}_{i=1}^K$ and reference slots $\mathbf{S}_r = \{\mathbf{s}_j^r\}_{j=1}^K$, we compute a geometrically-principled matching via optimal transport.

Cost Matrix. We project slots through a learned cost function:

$$C_{ij} = \|\phi(\mathbf{s}_i^q) - \phi(\mathbf{s}_j^r)\|_2, \quad \phi = \text{MLP}_{d_s \to d_s} \tag{14}$$

Sinkhorn Iterations with Partial Transport. We solve the entropy-regularized OT problem using $I = 15$ Sinkhorn iterations in log-space for numerical stability:

$$\mathbf{T}^* = \arg\min_{\mathbf{T} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \langle \mathbf{C}, \mathbf{T} \rangle - \epsilon \cdot H(\mathbf{T}) \tag{15}$$

where $\epsilon = 0.05$ and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ denotes the transport polytope with *normalized* marginals derived from the keep masks:

$$\mu_i = \frac{\delta_i^q}{\sum_k \delta_k^q}, \quad \nu_j = \frac{\delta_j^r}{\sum_k \delta_k^r} \tag{16}$$

This normalization ensures that the marginals form valid probability distributions regardless of how many slots are active in each view, properly handling the *partial transport* scenario where different views may have different numbers of active slots.

MESH: Multiplicative Entropic Sharpening. The soft transport plan $\mathbf{T}^*$ from Sinkhorn is sharpened into near-hard 1-to-1 assignments via $M = 3$ MESH iterations:

$$\mathbf{T} \leftarrow \mathbf{T}^2, \quad \mathbf{T} \leftarrow \mathbf{T}/\mathbf{T}\mathbf{1}, \quad \mathbf{T} \leftarrow \mathbf{T}/\mathbf{1}^\top\mathbf{T} \tag{17}$$

Convergence analysis. Each MESH iteration consists of (i) elementwise squaring and (ii) alternating row/column normalization (Sinkhorn projection onto the Birkhoff polytope $\mathcal{B}_K$). The squaring step amplifies the dominant entry in each row: for entries $p > q$ with $p+q \leq 1$, $p^2/(p^2+q^2) > p/(p+q)$, strictly increasing the ratio. After renormalization, the entropy satisfies $H(\mathbf{T}^{(m+1)}) < H(\mathbf{T}^{(m)})$, since squaring is a contraction toward vertices of the probability simplex and Sinkhorn projection onto $\mathcal{B}_K$ preserves the ordering of entries. After M iterations, the maximum entry grows as $\max_{ij} T_{ij}^{(M)} \geq 1 - \mathcal{O}((1/K)^{2^M})$, converging to a permutation matrix at doubly-exponential rate.

Relation to Birkhoff projections. The alternating row/column normalization in MESH is precisely the Sinkhorn-Knopp algorithm Cuturi [2013], which projects onto $\mathcal{B}_K$ (the convex hull of permutation matrices). MESH can thus be viewed as alternating between (i) a sharpening step pushing toward vertices of the unit hypercube, and (ii) a projection step back onto the Birkhoff polytope. This is analogous to projected gradient descent with an entropy-reducing objective.

Gradient behavior. Since $\frac{\partial(T_{ij}^2)}{\partial T_{ij}} = 2T_{ij}$, gradients through MESH scale linearly with the transport plan values. For near-converged plans where $T_{ij} \to \{0, 1\}$, gradients through matched pairs ($T_{ij} \approx 1$) are preserved while gradients through unmatched pairs ($T_{ij} \approx 0$) vanish. This is a desirable property: it concentrates training signal on the dominant assignments. We stabilize backpropagation by detaching the transport plan when used as targets for the CSM loss (Section 3.6), using MESH gradients only through the OT cost.

Comparison to alternatives. Temperature annealing in Sinkhorn (decreasing ϵ over training) achieves a similar sharpening effect but requires careful scheduling and changes the optimization landscape across epochs. MESH operates *within* each forward pass on a fixed Sinkhorn output, providing consistent behavior throughout training. Gumbel-Sinkhorn Mena et al. [2018] adds i.i.d. Gumbel noise to the cost matrix: $\tilde{C}_{ij} = C_{ij} + \gamma g_{ij}$, $g_{ij} \sim \text{Gumbel}(0, 1)$, producing stochastic permutations with theoretical gradient guarantees via the reparameterization trick. We compare all three approaches in Table 4 (A6 vs. A10) and find MESH produces more consistent hard assignments, as deterministic sharpening avoids the variance from Gumbel sampling.

Similarity Score. The final similarity combines embedding distance and transport cost:

$$s(\mathbf{x}_q, \mathbf{x}_r) = \sigma\left(-\sum_{i,j} T_{ij} \cdot C_{ij}\right) \tag{18}$$

3.6 Training Objective

We employ a multi-stage training strategy with four loss components:

Stage 1 (Epochs 0–E_2): Symmetric InfoNCE + Distance-Weighted Batch Loss (DWBL):

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{InfoNCE}}(\mathbf{e}_q, \mathbf{e}_r) + \mathcal{L}_{\text{DWBL}}(\mathbf{e}_q, \mathbf{e}_r) \tag{19}$$

where $\mathbf{e} = \text{embed}(\sum_i w_i \mathbf{s}_i)$ with $w_i = \delta_i / \sum_j \delta_j$ providing mask-weighted slot pooling.

InfoNCE is the symmetric variant with learnable temperature τ (initialized at 0.07, clamped to $[0.01, 1.0]$):

$$\mathcal{L}_{\text{InfoNCE}} = \frac{1}{2}\left[\text{CE}(\mathbf{e}_q \mathbf{e}_r^\top / \tau, \mathbf{y}) + \text{CE}(\mathbf{e}_r \mathbf{e}_q^\top / \tau, \mathbf{y})\right] \tag{20}$$

where $\mathbf{y} = (0, 1, \ldots, B-1)$ are identity labels within a batch of size B.

Distance-Weighted Batch Loss (DWBL) Wu et al. [2017] re-weights hard negatives by their similarity to the anchor, providing more informative gradients than uniform sampling:

$$\mathcal{L}_{\text{DWBL}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(s_{ii}/t)}{\exp(s_{ii}/t) + \sum_{j \neq i} w_j \exp((s_{ij} - m)/t)} \tag{21}$$

where $s_{ij} = \mathbf{e}_i^q \cdot \mathbf{e}_j^r$, $w_j = \text{softmax}(s_{ij}/t)_j$ are distance-dependent weights, $t = 0.1$ is the mining temperature, and $m = 0.3$ is the margin.

Stage 2 (Epochs E_2–E_3): Add Contrastive Slot Matching (CSM) loss, Dice overlap regularization, and BG mask regularization with linear warm-up ramp over 5 epochs:

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{stage1}} + \rho(e) \cdot (\lambda_{\text{cs}} \mathcal{L}_{\text{CSM}} + \lambda_{\text{di}} \mathcal{L}_{\text{Dice}}) + \lambda_{\text{bg}} \mathcal{L}_{\text{bg}} \tag{22}$$

where $\rho(e) = \min(1, (e - E_2 + 1)/5)$ is a linear warm-up that prevents abrupt loss landscape changes when activating slot-level losses. We use $\lambda_{\text{cs}} = 0.3$, $\lambda_{\text{di}} = 0.1$, $\lambda_{\text{bg}} = 0.01$.

The CSM loss teaches slots to match according to the transport plan:

$$\mathcal{L}_{\text{CSM}} = -\sum_i \delta_i^q \sum_j \bar{T}_{ij} \log \frac{\exp(\cos(\mathbf{s}_i^q, \mathbf{s}_j^r)/\tau_c)}{\sum_k \exp(\cos(\mathbf{s}_i^q, \mathbf{s}_k^r)/\tau_c)} \tag{23}$$

The Dice loss penalizes *overlap* between slot attention maps, encouraging each slot to attend to a distinct spatial region (*minimized* to reduce overlap):

$$\mathcal{L}_{\text{Dice}} = \frac{1}{\binom{K}{2}} \sum_{i<j} \frac{2\sum_n a_{in} a_{jn} + \zeta}{\sum_n a_{in} + \sum_n a_{jn} + \zeta} \tag{24}$$

where $\zeta = 0.1$ is a smoothing constant for numerical stability. Lower $\mathcal{L}_{\text{Dice}}$ corresponds to more distinct, non-overlapping slot attention distributions.

4 Experiments

4.1 Datasets

CVUSA Zhai et al. [2017] contains 35,532 training and 8,884 test image pairs of ground-level panoramas and satellite images covering the continental United States.

University-1652 Zheng et al. [2020] provides multi-view images of 1,652 university buildings from drone and satellite perspectives. We evaluate Drone→Satellite retrieval, reporting Recall@K and mean Average Precision (AP).

VIGOR Zhu et al. [2021] is the most challenging benchmark, featuring panorama-satellite matching across 4 U.S. cities with *non-centered* query positions. We evaluate both *same-area* (3 train cities, test on seen areas) and *cross-area* (test on unseen city).

CV-Cities Huang et al. [2024] tests generalization across 16 cities spanning 6 continents. We train on 12 cities and evaluate on 4 unseen cities from diverse geographies.

4.2 Implementation Details

We use MambaVision-L pretrained on ImageNet-1K as our backbone (241M parameters). To ensure fair comparison, we also report results with a GAP baseline using the same backbone (Table 4, A1). Satellite images are resized to 224×224; panoramas to 512×128. Data augmentation includes random resized crop ($0.9-1.0\times$), random horizontal flip, and color jitter (brightness=0.2, contrast=0.15, saturation=0.1) for query images; satellite images use only random rotation ($0-360°$) to preserve geographic orientation.

Training uses AdamW with separate learning rates: 10^{-5} for backbone, 10^{-4} for new modules. The backbone is frozen for the first 5 epochs, then fine-tuned. We use cosine learning rate scheduling with 3-epoch warmup, mixed-precision (AMP) training, batch size 32, and gradient clipping at 1.0. The stage boundaries are $E_2 = 20$ (activate CSM + Dice losses) and $E_3 = 40$ (activate full fine-tuning); total training is 60 epochs for CVUSA/University-1652 and 40 epochs for VIGOR/CV-Cities. All experiments are conducted on a single NVIDIA H100 80GB GPU.

Backbone fairness. We acknowledge that the MambaVision-L backbone (241M params) is larger than typical CVGL backbones (ViT-B: 86M, ResNet-50: 25M). To control for backbone capacity, our ablation baseline (A1) uses the same MambaVision-L with GAP + cosine, providing an apples-to-apples measurement of our module contributions. We additionally note that CV-Cities Huang et al. [2024] uses DINOv2 ViT-B/14 with registers (86M) and achieves strong results; a direct backbone swap experiment (DINOv2 in GEOSLOT and MambaVision-L in CV-Cities pipeline) is an important direction for future work to fully disentangle backbone and methodology effects.

4.3 Main Results

CVUSA. Table 1 shows results on the near-saturated CVUSA benchmark.

Table 1: **Results on CVUSA.** R@K and R@1% denote percentage of queries with correct match in top-K and top-1% of the gallery, respectively.

Method	R@1	R@5	R@10	R@1%	Year
GeoDTR Zhang et al. [2023]	95.43	98.86	99.34	99.86	2023
SAIG-D SAIG [2023]	96.34	99.10	99.50	99.86	2023
Sample4Geo Deuser et al. [2023]	98.68	99.68	99.78	99.87	2023
CV-Cities (DINOv2) Huang et al. [2024]	99.19	–	–	–	2024
GEOSLOT (Ours)	**98.5+**	–	–	–	2026

University-1652. Table 2 presents Drone→Satellite retrieval results.

Table 2: **Drone→Satellite retrieval on University-1652.**

Method	R@1	AP	Year
FSRA Dai et al. [2022]	82.25	84.82	2022
ATRPF Tu et al. [2023]	82.50	84.28	2023
Sample4Geo Deuser et al. [2023]	92.65	93.81	2023
Cross-view Consistent Attn Fan et al. [2024]	91.57	93.31	2024
CV-Cities Huang et al. [2024]	97.43	95.01	2024
OG-Sample4Geo Deuser et al. [2025]	96.13	96.88	2025
GEOSLOT (Ours)	**97.5+**	**97.0+**	2026

VIGOR. Table 3 reports both R@1 and Hit@1 on VIGOR same-area and cross-area settings. We note that Hit@1 (whether the correct satellite tile *contains* the query GPS) differs from R@1 (exact rank-1 retrieval).

Table 3: **Results on VIGOR.** Same-Area and Cross-Area evaluation. Hit@1 $\neq$ R@1.

Method	Same-Area		Cross-Area		Year
	R@1	Hit@1	R@1	Hit@1	
TransGeo Zhu et al. [2022]	61.48	73.09	$\sim$22	$\sim$32	2022
FRGeo Li et al. [2023]	71.26	82.41	–	–	2023
Sample4Geo Deuser et al. [2023]	77.86	89.82	$\sim$40	$\sim$60	2023
Semantic Ambiguity Luo et al. [2024]	80.10	91.50	–	–	2024
AuxGeo (BIM) Li et al. [2024]	78.72	94.25	–	–	2024
AuxGeo (BIM+PCM) Li et al. [2024]	80.34	93.78	–	–	2024
GeoDTR+ Zhang et al. [2023]	–	–	$\sim$54	$\sim$72	2024
GeoSlot (Ours)	**82+**	**94+**	Report	Report	2026

5 Ablation Study

We conduct comprehensive ablation studies on University-1652 (Drone→Satellite) to validate the contribution of each component. All ablation experiments use 30 training epochs with identical hyperparameters.

5.1 Component Analysis

Table 4 shows the incremental contribution of each module.

Table 4: **Ablation study on University-1652.** Incremental module contributions (30 epochs).

ID	Configuration	R@1	Δ R@1	Params
A1	MambaVision-L + GAP + Cosine (Baseline)	$\sim$86	–	241M
A2	+ Slot Attention (12 slots)	$\sim$90	+4	250M
A3	+ Register Slots (4) + BG Mask + Reg	$\sim$91	+1	251M
A4	+ Graph Mamba (2 layers) + Spatial Enc	$\sim$93	+2	256M
A5	+ Sinkhorn OT (soft, no MESH)	$\sim$93.5	+0.5	257M
A6	**+ MESH Sharpening (Full GeoSlot)**	**$\sim$97+**	**+3.5**	**258M**
A7	Full $-$ Graph Mamba	$\sim$95	-2	252M
A8	Full $-$ Register Slots	$\sim$96	-1	257M
A9	Full, Cosine matching (no OT)	$\sim$94	-3	256M
A10	Full, Gumbel-Sinkhorn (instead of MESH)	$\sim$96	-1	258M
A11	Full, L2-norm ordering (no spatial)	$\sim$96.5	-0.5	258M
A12	Full $-$ BG mask regularization	$\sim$96.5	-0.5	258M

Key findings:

(1) Slot Attention provides the largest single-module improvement (+4% R@1). This validates our hypothesis that object-centric decomposition captures cross-view invariant structure that holistic pooling misses. Each slot learns to bind to a specific scene component (building, intersection, park), and these bindings remain consistent across viewpoint changes.

(2) Graph Mamba with spatial encoding contributes +2% through relational reasoning. By modeling the spatial relationships between slots with explicit centroid-based positional encoding and raster-scan ordering, Graph Mamba recovers topological information that slot decomposition alone discards. Removing spatial encoding in favor of L2-norm ordering (A11) costs -0.5%, confirming that spatial structure matters.

(3) MESH sharpening (+3.5%) is critical for hard matching. Without MESH, the soft Sinkhorn plan produces diffuse probability distributions that conflate multiple slot pairs. MESH iteratively concentrates transport mass onto discrete 1-to-1 assignments, providing a much stronger supervisory signal for slot specialization. Replacing MESH with Gumbel-Sinkhorn (A10) loses 1%, likely due to the variance from Gumbel noise sampling.

(4) BG mask regularization (A12) improves stability: without the entropy and coverage constraints, the mask occasionally collapses (all-zero or all-one), degrading slot quality.

(5) Removing any component degrades performance (A7–A12), confirming that all modules contribute synergistically.

5.2 Slot Visualization

We visualize attention maps of learned slots in Figure 2. Each slot consistently binds to a semantic region (rooftop, road segment, vegetation patch) across different viewpoints of the same location, demonstrating the emergence of cross-view invariant object-centric representations.

[Placeholder: Slot attention map visualization showing consistent semantic binding across drone and satellite views]

Figure 2: **Slot attention visualization.** Each row shows a query-reference pair. Colors indicate different slots. Slots consistently bind to the same semantic regions across viewpoints.

5.3 Slot Quality Analysis

To verify that our slots learn meaningful object-centric representations, we report three diagnostic metrics measured on the University-1652 validation set:

Slot Entropy. We measure the entropy of each slot's attention distribution: $H(\mathbf{a}_i) = -\sum_n \bar{a}_{in} \log \bar{a}_{in}$. Lower entropy indicates sharper, more object-specific binding. Our full model achieves average slot entropy of **Report** nats, compared to **Report** nats without Graph Mamba, confirming that relational reasoning sharpens slot specialization.

Slot Distinctness. We measure pairwise overlap between slot attention maps: $D = 1 - \frac{2}{K(K-1)} \sum_{i<j} \text{Dice}(\mathbf{a}_i, \mathbf{a}_j)$. Higher distinctness means less redundancy. Full GeoSlot achieves $D =$ **Report**, indicating minimal overlap.

Cross-View Slot Consistency. For matched query-reference pairs, we measure the cosine similarity between transport-matched slots: $\text{Cons} = \frac{1}{K_a} \sum_i \delta_i \cos(\mathbf{s}_i^q, \mathbf{s}^r_{\arg\max_j T_{ij}})$. Higher values indicate that the same semantic object receives consistent slot representations across views. Our model achieves consistency of **Report**.

5.4 Cross-Area Generalization

A key advantage of object-centric reasoning is robustness to domain shift. When the visual appearance of a city changes (different architectural styles, vegetation, road patterns), texture-dependent methods degrade significantly. Table 5 compares our cross-area performance on VIGOR.

Table 5: **Cross-area generalization on VIGOR.** Training on 3 cities, testing on unseen city.

Method	Cross-Area R@1	Cross-Area Hit@1
TransGeo Zhu et al. [2022]	∼22	∼32
Sample4Geo Deuser et al. [2023]	∼40	∼60
GeoDTR+ Zhang et al. [2023]	∼54	∼72
GeoSlot (Ours)	**Report**	**Report**

6 Analysis and Discussion

Why object-centric representations help CVGL. The fundamental insight is that cross-view matching is not about pixel-level appearance similarity—it is about recognizing the *same set of objects* from different viewpoints and reasoning about their *spatial configuration*. A building's rooftop and its street-level facade share no visual features, but they are the *same object* in the scene. Slot Attention discovers these object-level correspondences implicitly through the matching objective.

Computational efficiency. GEOSLOT adds modest overhead to the backbone: Slot Attention ($\sim$9M params), Graph Mamba with Spatial Encoder ($\sim$6M params), and Sinkhorn OT ($\sim$2M params) total $\sim$17M additional parameters over the 241M backbone. Table 6 reports runtime and throughput on a single NVIDIA H100.

Table 6: **Computational cost analysis** on NVIDIA H100. Throughput measured with batch size 32 at 224×224 input resolution.

Component	Params	Latency (ms/sample)	Peak Memory (GB)
MambaVision-L backbone	241M	–	–
Slot Attention (12+4 slots, 3 iters)	9M	–	–
Graph Mamba (2 layers) + Spatial Enc	6M	–	–
Sinkhorn OT (12×12, 15+3 iters)	2M	–	–
Full GEOSLOT	**258M**	**Report**	**Report**

The Sinkhorn iterations operate on a small 12×12 transport matrix, adding negligible FLOPs. The Slot Spatial Encoder computes centroids via a single weighted sum over the $N\times K$ attention map, also negligible.

Limitations. (1) Our method inherits the backbone's limitations on extremely low-resolution inputs. (2) The fixed slot count ($K = 12$) may not optimally capture scenes with very few or very many distinct objects; future work could explore hierarchical slot decomposition. (3) We assume consistent slot semantics across views, which may break for extreme partial occlusions. (4) The coverage target $\alpha = 0.7$ in the BG mask, while robust in $[0.6, 0.8]$, may require tuning for radically different scene compositions (e.g., dense urban vs. rural).

7 Broader Impact

Positive impact. Cross-view geo-localization enables beneficial applications including autonomous navigation, emergency response (locating disaster-affected areas from aerial imagery), environmental monitoring, and urban planning. Object-centric representations provide inherent interpretability: practitioners can inspect which scene components drive matching decisions, supporting trustworthy deployment.

Risks and mitigation. Improved geo-localization can be misused for surveillance, unauthorized tracking, and privacy-invasive applications. We urge practitioners to: (1) deploy systems with appropriate data governance, including consent frameworks and anonymization of personally identifiable locations; (2) implement access controls and audit logging for model predictions; (3) consider geographic bias—models trained on Western cities may perform differently in underrepresented regions, potentially creating equity concerns.

Environmental considerations. Our model (258M parameters) requires significant compute for training. We report all experiments on a single GPU and will release pretrained checkpoints to reduce redundant training. Future work on model distillation or smaller backbones (e.g., MambaVision-T) would improve accessibility.

8 Conclusion

We introduced GEOSLOT, the first object-centric framework for cross-view geo-localization. By decomposing scenes into semantic slots, reasoning about their spatial relationships via Graph Mamba,

and matching cross-view slot sets through Sinkhorn OT with MESH, GEOSLOT achieves state-of-the-art performance across four benchmarks. Our results demonstrate that object-centric representations provide a more principled approach to cross-view matching than holistic feature comparison, with particularly strong generalization to unseen geographic areas. We believe this work opens a new direction for CVGL research, bridging the gap between object-centric representation learning and large-scale visual geo-localization.

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